

Foundations of machine learning
Experiments for policy choice

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Outline

- Alternative objectives for the design of experiments.
- Exploration sampling as a modification of Thompson sampling.
- The oracle optimal allocation for the policy choice problem.
- Exploration sampling converges to the oracle optimal allocation.
- Simulations and empirical application.

Takeaways for this part of class

- Adaptive designs improve expected welfare.
- Features of the optimal treatment assignment:
 - Shift toward better performing treatments over time.
 - But don't shift as much as for Bandit problems:
We have no "exploitation" motive!
 - Asymptotically: Equalize power for comparisons of each suboptimal treatment to the optimal one.
- Fully optimal assignment is computationally challenging in large samples.
- We propose a simple **exploration sampling** algorithm.
 - Argue that it is rate-optimal for our problem, because it equalizes power across suboptimal treatments.
 - Show that it dominates alternatives in calibrated simulations.

Introduction

The goal of many experiments is to inform policy choices:

1. **Job search assistance** for refugees:

- Treatments: Information, incentives, counseling, ...
- Goal: Find a policy that helps as many refugees as possible to find a job.

2. **Clinical trials**:

- Treatments: Alternative drugs, surgery, ...
- Goal: Find the treatment that maximizes the survival rate of patients.

3. Online **A/B testing**:

- Treatments: Website layout, design, search filtering, ...
- Goal: Find the design that maximizes purchases or clicks.

4. Testing **product design**:

- Treatments: Various alternative designs of a product.
- Goal: Find the best design in terms of user willingness to pay.

What is the objective of your experiment?

1. Getting precise treatment effect estimators, powerful tests:

$$\min \sum_d (\hat{\theta}^d - \theta^d)^2$$

⇒ Standard experimental design recommendations.

2. Maximizing the outcomes of experimental participants:

$$\max \sum_i \theta^{D_i}$$

⇒ Multi-armed bandit problems.

3. Picking a welfare maximizing policy after the experiment:

$$\max \theta^{d^*},$$

where d^* is chosen after the experiment.

⇒ This lecture.

Setup

Thompson sampling and exploration sampling

Calibrated simulations

Implementation in the field

References

Setup

- Waves $t = 1, \dots, T$, sample sizes N_t .
- Treatment $D \in \{1, \dots, k\}$, outcomes $Y \in \{0, 1\}$.
- Potential outcomes Y^d .
- Repeated cross-sections:
($Y_{it}^0, \dots, Y_{it}^k$) are i.i.d. across both i and t .
- Average potential outcome:
$$\theta^d = E[Y_{it}^d].$$
- Key choice variable:
Number of units n_t^d assigned to $D = d$ in wave t .
- Outcomes:
Number of units s_t^d having a “success” (outcome $Y = 1$).

Treatment assignment, outcomes, state space

- Treatment assignment in wave t : $\mathbf{n}_t = (n_t^1, \dots, n_t^k)$.
- Outcomes of wave t : $\mathbf{s}_t = (s_t^1, \dots, s_t^k)$.
- Cumulative versions:

$$M_t = \sum_{t' \leq t} N_{t'},$$

$$\mathbf{m}_t = \sum_{t' \leq t} \mathbf{n}_{t'},$$

$$\mathbf{r}_t = \sum_{t' \leq t} \mathbf{s}_{t'}.$$

- Relevant information for the experimenter in period $t+1$ is summarized by $\mathbf{m}_t$ and $\mathbf{r}_t$.
- Total trials for each treatment, total successes.

Design objective and Bayesian prior

- **Policy objective** $\theta^{d_T^*}$.
 - where d_T^* is chosen after the experiment.
- **Prior**
 - $\theta^d \sim \text{Beta}(\alpha_0^d, \beta_0^d)$, independent across d .
 - Posterior after period t : $\theta^d | \mathbf{m}_t, \mathbf{r}_t \sim \text{Beta}(\alpha_t^d, \beta_t^d)$
$$\alpha_t^d = \alpha_0^d + r_t^d$$
$$\beta_t^d = \beta_0^d + m_t^d - r_t^d.$$

- **Posterior expected social welfare**
as a function of d :

$$\begin{aligned} SW_T(d) &= E[\theta^d | \mathbf{m}_T, \mathbf{r}_T], \\ &= \frac{\alpha_T^d}{\alpha_T^d + \beta_T^d}, \\ d_T^* &\in \operatorname{argmax}_d SW_T(d). \end{aligned}$$

Regret

- True optimal treatment: $d^{(1)} \in \arg \max_{d'} \theta^{d'}$.
- **Policy regret** when choosing treatment d :

$$\Delta^d = \theta^{d^{(1)}} - \theta^d.$$

- Maximizing expected social welfare is equivalent to minimizing the expected policy regret at T ,

$$E[\Delta^d | \mathbf{m}_T, \mathbf{r}_T] = \theta^{d^{(1)}} - SW_T(d)$$

- **In-sample regret**: Objective considered in the bandit literature,

$$\frac{1}{M} \sum_{i,t} \Delta^{D_{it}}.$$

Different from policy regret $\Delta^{d_T^*}$!

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Reminder: Thompson sampling

- **Thompson sampling**

Assign each treatment with probability equal to the posterior probability that it is optimal.

$$p_t^d = P \left(d = \operatorname{argmax}_{d'} \theta^{d'} | \mathbf{m}_{t-1}, \mathbf{r}_{t-1} \right).$$

- Easily implemented: Sample draws $\hat{\theta}_{it}$ from the posterior, assign

$$D_{it} = \operatorname{argmax}_d \hat{\theta}_{it}^d.$$

- **Expected Thompson sampling**

- Straightforward modification for the batched setting.
- Assign non-random shares p_t^d of each wave to treatment d .

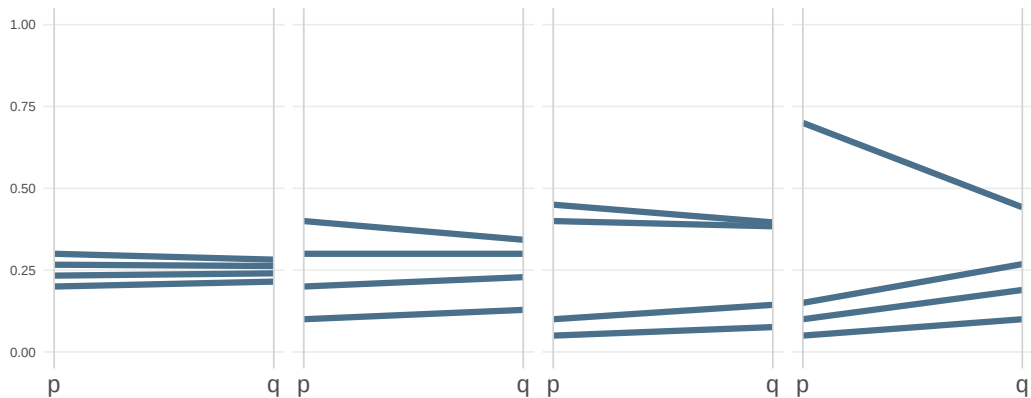
Exploration sampling

- Agrawal and Goyal (2012) proved that Thompson-sampling is rate-optimal for the multi-armed bandit problem.
- It is not for our policy choice problem!
- We propose the following modification.
- **Exploration sampling:**
Assign shares q_t^d of each wave to treatment d , where

$$q_t^d = S_t \cdot p_t^d \cdot (1 - p_t^d),$$
$$S_t = \frac{1}{\sum_d p_t^d \cdot (1 - p_t^d)}.$$

- This modification
 1. yields rate-optimality, and
 2. improves performance in our simulations.

Illustration of the mapping from Thompson to exploration sampling



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Calibrated simulations

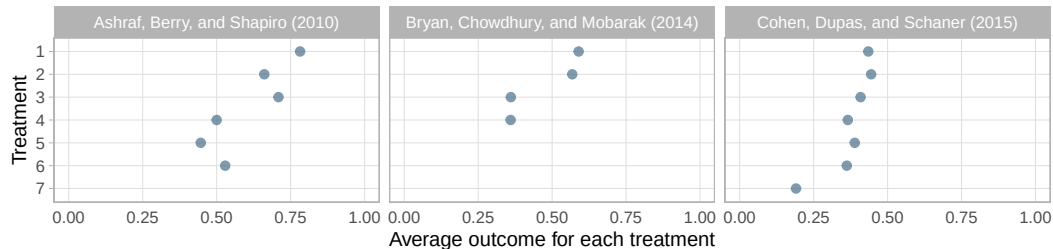
- Simulate data calibrated to estimates of 3 published experiments.
- Set θ equal to observed average outcomes for each stratum and treatment.
- Total sample size same as original.

Ashraf, N., Berry, J., and Shapiro, J. M. (2010). Can higher prices stimulate product use? Evidence from a field experiment in Zambia. *American Economic Review*, 100(5):2383–2413

Bryan, G., Chowdhury, S., and Mobarak, A. M. (2014). Underinvestment in a profitable technology: The case of seasonal migration in Bangladesh. *Econometrica*, 82(5):1671–1748

Cohen, J., Dupas, P., and Schaner, S. (2015). Price subsidies, diagnostic tests, and targeting of malaria treatment: evidence from a randomized controlled trial. *American Economic Review*, 105(2):609–45

Calibrated parameter values



Treatment arms labeled 1 up to 7:

- Ashraf et al. (2010): Kw 300 - 800 price for water disinfectant.
- Bryan et al. (2014): Migration incentives - cash, credit, information, and control.
- Cohen et al. (2015): Price of Ksh 40, 60, and 100 for malaria tablets, each with and without free malaria test, and control of Ksh 500.

Summary of simulation findings

- With two waves, relative to non-adaptive assignment:
 - Thompson reduces average policy regret by 15-58 %,
 - exploration sampling by 21-67 %.
- Similar pattern for the probability of choosing the optimal treatment.
- Gains increase with the number of waves, given total sample size.
 - Up to 85% for exploration sampling with 10 waves for Ashraf et al. (2010).
- Gains largest for Ashraf et al. (2010), followed by Cohen et al. (2015), and smallest for Bryan et al. (2014).
- For in-sample regret, Thompson is best, followed closely by exploration sampling.

Ashraf, Berry, and Shapiro (2010)

Statistic	2 waves	4 waves	10 waves
Average policy regret			
exploration sampling	0.0017	0.0010	0.0008
expected Thompson	0.0022	0.0014	0.0013
non-adaptive	0.0051	0.0050	0.0051
Share optimal			
exploration sampling	0.978	0.987	0.989
expected Thompson	0.971	0.981	0.982
non-adaptive	0.933	0.935	0.933
Average in-sample regret			
exploration sampling	0.1126	0.0828	0.0701
expected Thompson	0.1007	0.0617	0.0416
non-adaptive	0.1776	0.1776	0.1776
Units per wave	502	251	100

Bryan, Chowdhury, and Mobarak (2014)

Statistic	2 waves	4 waves	10 waves
Average policy regret			
exploration sampling	0.0045	0.0041	0.0039
expected Thompson	0.0048	0.0044	0.0043
non-adaptive	0.0055	0.0054	0.0054
Share optimal			
exploration sampling	0.792	0.812	0.820
expected Thompson	0.777	0.795	0.801
non-adaptive	0.747	0.748	0.749
Average in-sample regret			
exploration sampling	0.0655	0.0386	0.0254
expected Thompson	0.0641	0.0359	0.0205
non-adaptive	0.1201	0.1201	0.1201
Units per wave	935	467	187

Cohen, Dupas, and Schaner (2015)

Statistic	2 waves	4 waves	10 waves
Average policy regret			
exploration sampling	0.0070	0.0063	0.0060
expected Thompson	0.0074	0.0065	0.0061
non-adaptive	0.0086	0.0087	0.0085
Share optimal			
exploration sampling	0.567	0.586	0.592
expected Thompson	0.560	0.582	0.589
non-adaptive	0.526	0.524	0.529
Average in-sample regret			
exploration sampling	0.0489	0.0374	0.0314
expected Thompson	0.0467	0.0345	0.0278
non-adaptive	0.0737	0.0737	0.0737
Units per wave	1080	540	216

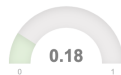
Implementation in the field

- NGO Precision Agriculture for Development (PAD), and Government of Odisha, India.
- Enrolling rice farmers into customized advice service by mobile phone.
- Waves of 600 farmers called through automated service; total of 10K calls.
- Outcome: did the respondent answer the enrollment questions?

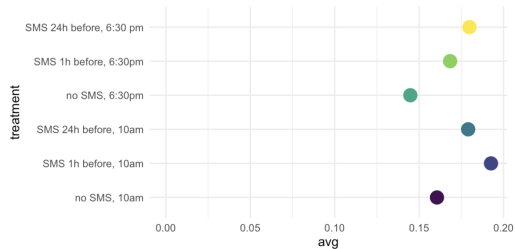
10000

Number of observations

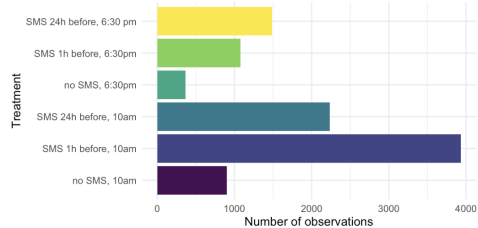
Success rate



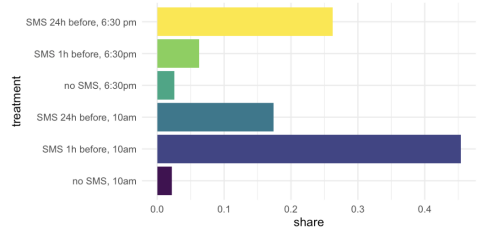
Success rate by treatment



Past distribution across treatments



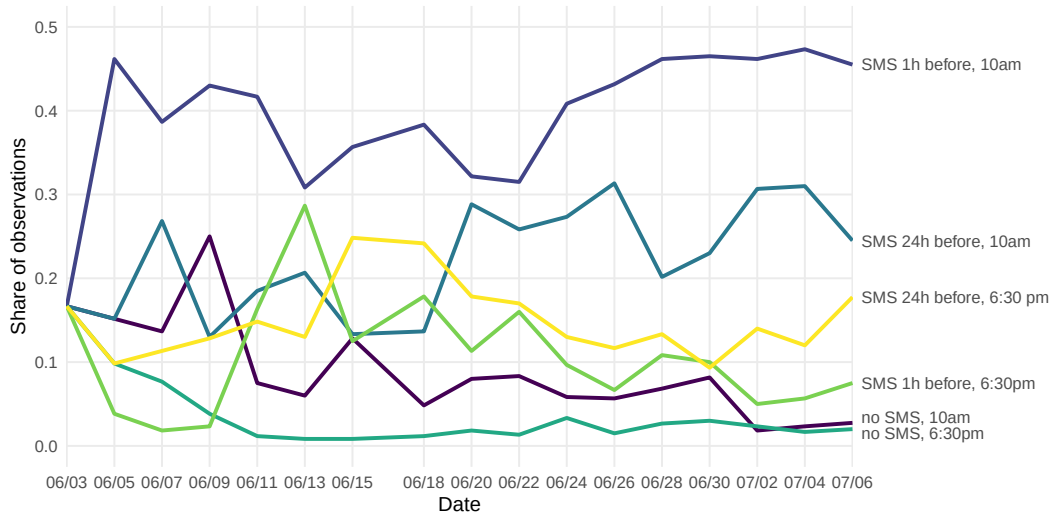
Current assignment probabilities



Outcomes and posterior parameters

Treatment		Outcomes			Posterior		
Call time	SMS alert	m_T^d	r_T^d	r_T^d/m_T^d	mean	SD	p_T^d
10am	-	903	145	0.161	0.161	0.012	0.009
10am	1h ahead	3931	757	0.193	0.193	0.006	0.754
10am	24h ahead	2234	400	0.179	0.179	0.008	0.073
6:30pm	-	366	53	0.145	0.147	0.018	0.011
6:30pm	1h ahead	1081	182	0.168	0.169	0.011	0.027
6:30 pm	24h ahead	1485	267	0.180	0.180	0.010	0.126

Assignment shares over time



References

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- Russo, D. (2016). *Simple bayesian algorithms for best arm identification*. In Conference on Learning Theory, pages 1417–1418.
- Kasy, M. and Sautmann, A. (2021). *Adaptive treatment assignment in experiments for policy choice*. Econometrica, 89(1):113–132.
- Interactive dashboard for treatment assignment:
https://maxkasy.shinyapps.io/exploration_sampling_dashboard/